Day 03: Regression with `scikit-learn`

Predicting continuous values

MSU REU Machine Learning Short Course

Learning Goals for Today

By the end of this activity, you'll be able to:

1. **Distinguish regression from classification** — when each is the right tool
2. **Build and train a linear regression model** using the same pipeline as classification
3. **Evaluate regression models** with MSE, R^2, and diagnostic residual plots
4. **Diagnose what your model is missing** by reading residuals
5. **Compare different regression models** (Linear vs Random Forest)
6. **Improve a model** by adding features and per-class approaches

This is the same workflow as classification. The only difference: you're predicting a number, not a category.

The Real Difference: One Question

	Classification	Regression
Question	"What is this?"	"How much?"
Output	A category (STAR, GALAXY, QSO)	A number (z-magnitude, temperature)
Example	Classify a spectrum as a star or quasar	Predict the brightness of that star

Same dataset. Same features. Different target variable.

Why Regression? Real Examples

Beyond classification, we often need **continuous predictions**:

Problem	Features	Target
Stellar brightness	Colors (u, g, r, i, z)	Magnitude (0–30)
Molecular energy	Atomic structure	Bond energy (continuous)
Star temperature	Spectral absorption lines	Temperature (K)
Galaxy redshift	Brightness + colors	Redshift z (0–7)

Key insight: When you need a precise number, not just a category, regression is your tool.

Linear Regression: The Simplest Model

The idea

Fit a straight line (or plane, or hyperplane) through the data.

Prediction formula:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

Where:

- $\hat{y}$ = predicted value
- x_i = features (colors, spectrum)
- β_i = learned weights (how much each feature matters)

Goal: Find coefficients **β** that minimize error.

Measuring Error in Regression

What is a residual?

$$\text{residual} = \text{predicted} - \text{actual}$$

Large residual = bad prediction. Small residual = good prediction.

Why minimize squared error (MSE)?

- Penalizes big mistakes harder than small ones
- Mathematically convenient (smooth, differentiable)
- Interpretable: MSE is in squared units of your target

Example: If predicting magnitude (0–30), an error of 5 is worse than 5 errors of 1.

The Regression Pipeline

Identical to classification: Prepare → Scale → Train → Predict → Evaluate

The only difference: your target is continuous, not categorical.

Building the Model (Still the Same Pattern)

The `scikit-learn` workflow is **identical to KNN**:

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# 1. Prepare: split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# 2. Scale: fit on training data only
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Evaluation: MSE and R^2

Mean Squared Error (MSE) — average squared distance between predicted and actual

$$\mathrm{MSE} = \frac{1}{n}\sum(y_{\text{true}} - y_{\text{pred}})^2$$

Lower is better. In same units as your squared target.

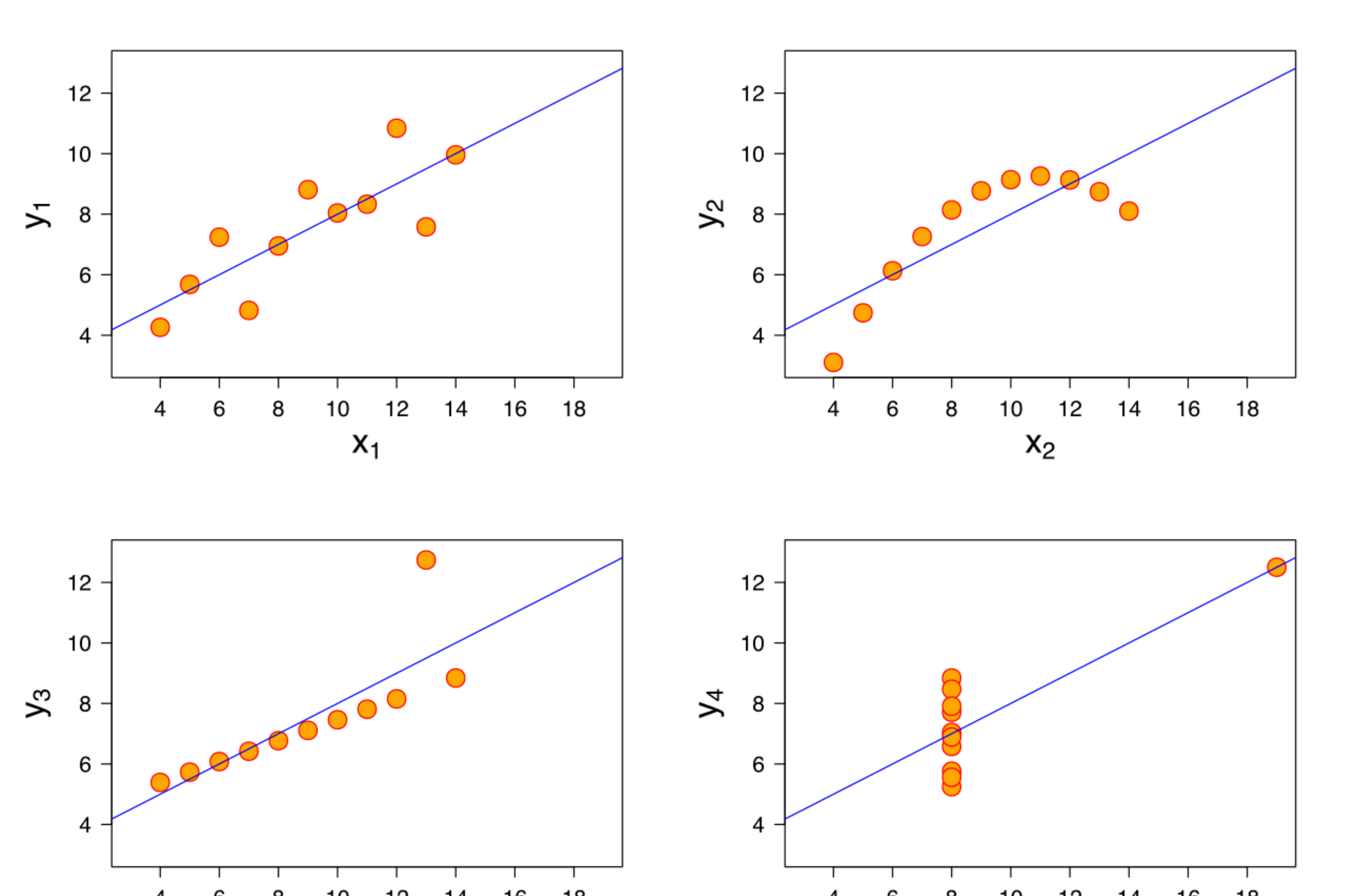
R^2 (R-squared) — fraction of variance explained

$$R^2 = 1 - \frac{\mathrm{SS}_{\text{res}}}{\mathrm{SS}_{\text{tot}}}$$

- 1.0 = perfect fit
- 0.0 = no better than predicting the mean
- 0.95 = good, but always verify with plots

A Warning: Anscombe's Quartet

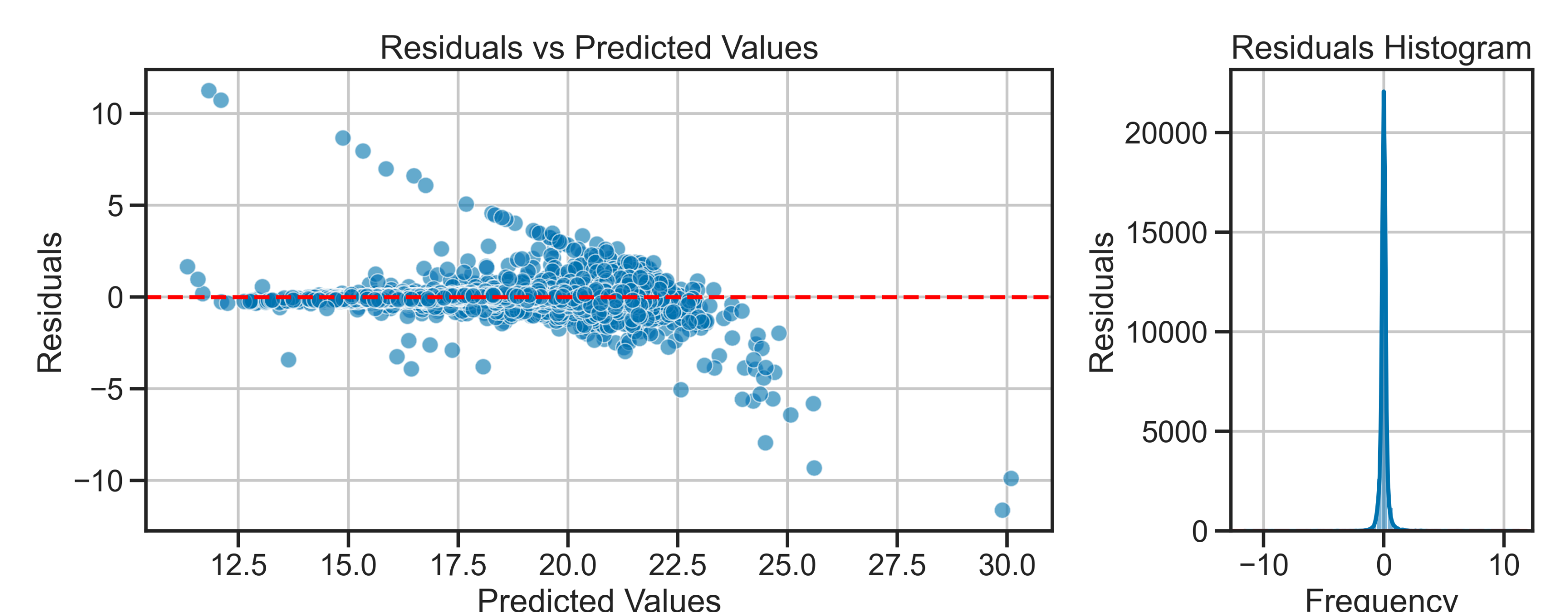
Four datasets. Identical means, variances, correlations, AND R^2 values.



Diagnostic Plots: Reading Residuals

Always visualize three things:

1. **Residuals vs Predicted** — scatter plot: do residuals look random?
2. **Histogram of residuals** — distribution: is it centered at zero?
3. **Actual vs Predicted** — 45° line should align with your points



What to Look For in Residuals

Rule: Residuals should look like **random noise** around zero.

If they have a pattern (curved, fanned, clustered), your model is missing something.

When Residuals Tell You Something

Your model doesn't fit

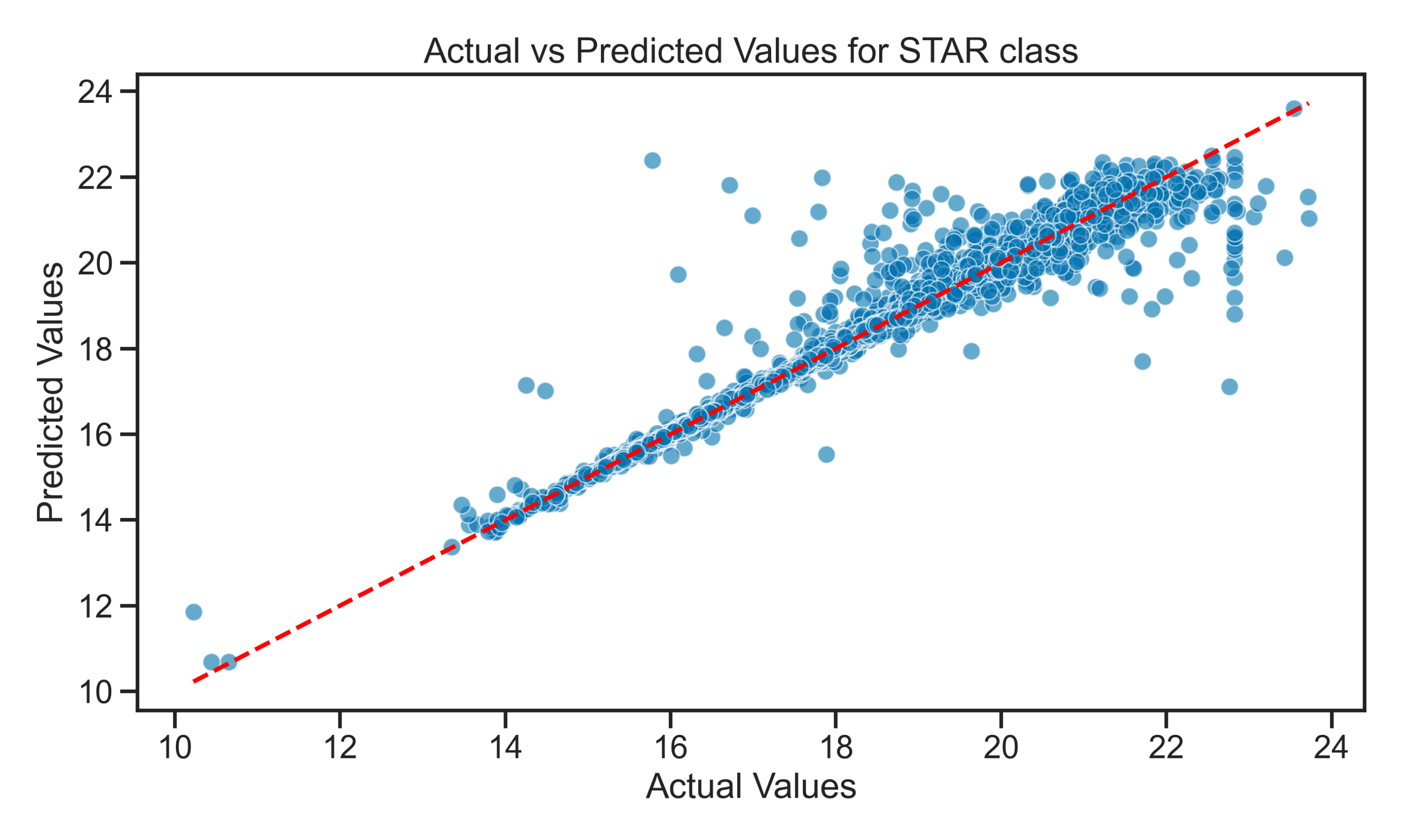
- **Curved pattern** → Relationship is non-linear (not a straight line)
- **Fan shape** → Error gets bigger at high values (heteroscedasticity)
- **Clustering** → Model works better for some groups than others

What to do?

- Try a **different model** (Random Forest, polynomial)
- **Add more features** (maybe colors interact?)
- **Separate by class** (STAR and GALAXY have different color-brightness relationships)

When Linear Isn't Enough

Same data, same question. Different model.



Model Comparison: Linear vs Random Forest

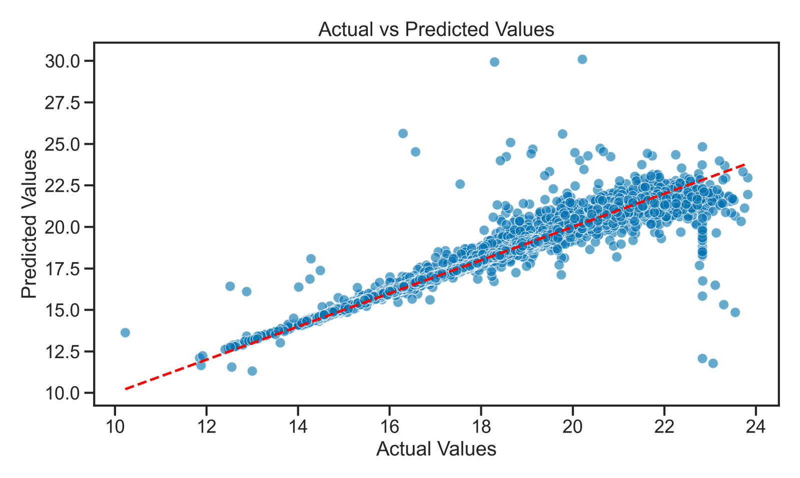
Linear regression assumes a straight line. Fast, interpretable, fails on curves.

Random Forest is flexible but slower and prone to overfitting.

Which to use? Visualize your residuals first. If they're curved, try Random Forest. If they're random, Linear might be enough.

Does Redshift Dominate Here Too?

Parallel to classification: does one feature carry all the signal?



Without redshift	With redshift
Low R^2 (colors alone are weak predictors)	High R^2 (redshift correlates with magnitude)
Colors matter	But is it "cheating"?

Key question: Adding redshift improves R^2, but does the model learn color-magnitude relationships, or just memorize redshift?

Per-Class Regression: Better than One-Size-Fits-All

Stars, galaxies, and quasars have different color-magnitude relationships.

Brighter stars → different color changes than brighter galaxies.

```
# Naive: one model for all 100,000 objects
model = LinearRegression()
model.fit(X_train, y_train)

# Better: fit one model per class
models = {}
for class_name in ["STAR", "GALAXY", "QSO"]:
    X_class = X_train[y_class_train == class_name]
    y_class = y_train[y_class_train == class_name]
```

Today's Activity: Regression with scikit-learn

Work through **Activity 03** step by step:

Part 1: Baseline Linear Model

- Load SDSS data (100k objects, 17 features)
- Predict **z-band magnitude** from colors (u, g, r, i)
- Evaluate with **MSE, R^2**

Part 2: Diagnose with Residuals

- Plot residuals vs predicted
- Is there a pattern? Curved? Fanned out?
- What does that tell you?

Part 3: Add Information